

Mean–Variance Portfolio Optimization with Monte Carlo Simulation and Scenario-Based Frontier Analysis

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Abstract

This paper implements a classical mean–variance portfolio optimization framework and augments it with Monte Carlo exploration, short-selling policy comparison, and a sample average approximation (SAA) study of parameter uncertainty. Two scenario models—*AI_Infra_Balanced* and *Macro_Diversified*—are specified by expected returns, volatilities, and correlation matrices. We compute efficient frontiers via a closed-form KKT solution under the full-investment constraint, visualize random portfolio opportunity sets under long-only and shorts-allowed policies, and compare the efficient frontier derived from true parameters to one estimated by SAA. Results show stable, near-collinear frontiers across a wide risk-aversion sweep and highlight risk amplification from unconstrained shorting. Figures are included in the Appendix.

1. Problem Description

We study the allocation of wealth across n risky assets with expected excess return vector $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$. The investor solves the budget-constrained mean–variance program

$$\begin{aligned} \min_{w \in \mathbb{R}^n} \quad & \frac{\gamma}{2} w^\top \Sigma w - \mu^\top w \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1 \end{aligned} \tag{1}$$

where $\gamma > 0$ is the risk-aversion parameter. We analyze two investment policies:

- **Long-only:** $w \geq 0$, $\mathbf{1}^\top w = 1$ (explored via Monte Carlo sampling).
- **Shorts-OK:** no sign constraints, $\mathbf{1}^\top w = 1$ (optimized via KKT closed form).

Our objectives are: (i) visualize opportunity sets from random weights; (ii) construct efficient frontiers by sweeping γ ; (iii) quantify parameter uncertainty using SAA.

2. Data Preparation and Pipeline

We specify two scenarios:

- **AI_Infra_Balanced:** assets represent AI-related equity, U.S. core bonds, REITs, and gold with moderate cross-asset diversification and defensive sleeves.

- **Macro_Diversified:** assets represent U.S./international/emerging equities, TIPS, and commodities to capture macro inflation and growth exposures.

Inputs are the vectors of expected annual returns μ , volatilities σ , and correlation matrices R . We construct $\Sigma = \text{diag}(\sigma) R \text{diag}(\sigma)$ and ensure positive definiteness via a minimal diagonal jitter if required. For Monte Carlo clouds we draw long-only weights from a Dirichlet distribution (nonnegative, sum-to-one) and shorts-allowed weights by normalizing zero-mean Gaussian draws to satisfy $\mathbf{1}^\top w = 1$.

3. Research Design

We evaluate:

- (a) **Opportunity sets:** risk–return clouds under long-only vs. shorts-allowed sampling.
- (b) **Efficient frontier:** for a grid $\Gamma = \{\gamma_k\}$, solve (1) to obtain $(\sigma(w_{\gamma_k}), \mu^\top w_{\gamma_k})$.
- (c) **SAA frontier:** simulate i.i.d. multivariate-normal returns of size N , compute $(\hat{\mu}, \hat{\Sigma})$, then repeat the frontier construction to assess estimation effects.

We report concise qualitative comparisons in the main text; all figures appear in the Appendix.

4. Programming and Implementation

Implementation is in Python (NumPy/Pandas/Matplotlib). Key components:

- **Covariance builder:** $\Sigma = DRD$ with $D = \text{diag}(\sigma)$; a nearest-SPD routine adds minimal jitter for Cholesky feasibility.
- **KKT closed form:** augment $\gamma\Sigma$ with the budget constraint to solve a $(n+1) \times (n+1)$ linear system and recover w for each γ .
- **Sampling:** long-only via Dirichlet; shorts-allowed via normalized Gaussian draws; vectorized portfolio mean and variance computations.
- **Reproducibility:** deterministic seeds; exported CSVs for frontiers and clouds; figures saved as PNG.

5. Results and Discussion

Across both scenarios, efficient frontiers are smooth and nearly linear in the $(\sigma, \mathbb{E}[r])$ plane over the tested γ range. The SAA frontier closely tracks the true-parameter frontier, with small level shifts consistent with finite-sample estimation. Long-only clouds exhibit a compact arch shape; allowing shorts expands the feasible region substantially and introduces extreme high-variance points, reflecting leverage and offsetting exposures. The *AI_Infra_Balanced* scenario shows higher attainable returns due to an equity-heavy sleeve, while *Macro_Diversified* displays broader cross-asset diversification and a slightly flatter risk–return trade-off. For practical allocation, imposing nonnegativity and/or leverage limits is advisable to avoid tail-risk outliers observed under unconstrained shorting.

Conclusion

This study integrates classical mean–variance optimization with Monte Carlo exploration and SAA-based robustness checks. The closed-form KKT solver provides an efficient way to trace the frontier, while sampling illuminates the feasible universe and policy consequences. Extensions include long-only quadratic programming, turnover and leverage constraints, and alternative risk measures (e.g., CVaR).

Appendix: Figures

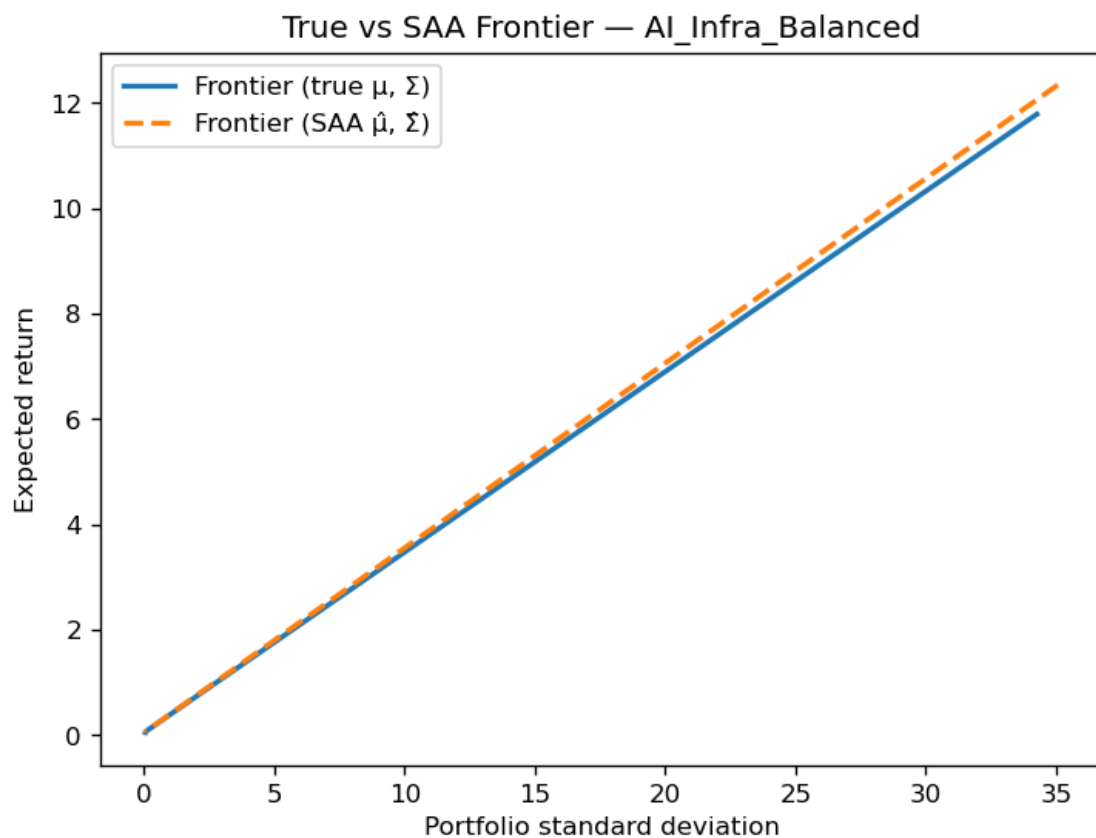


Figure 1: Efficient Frontier (True vs. SAA) — AI_Infra_Balanced

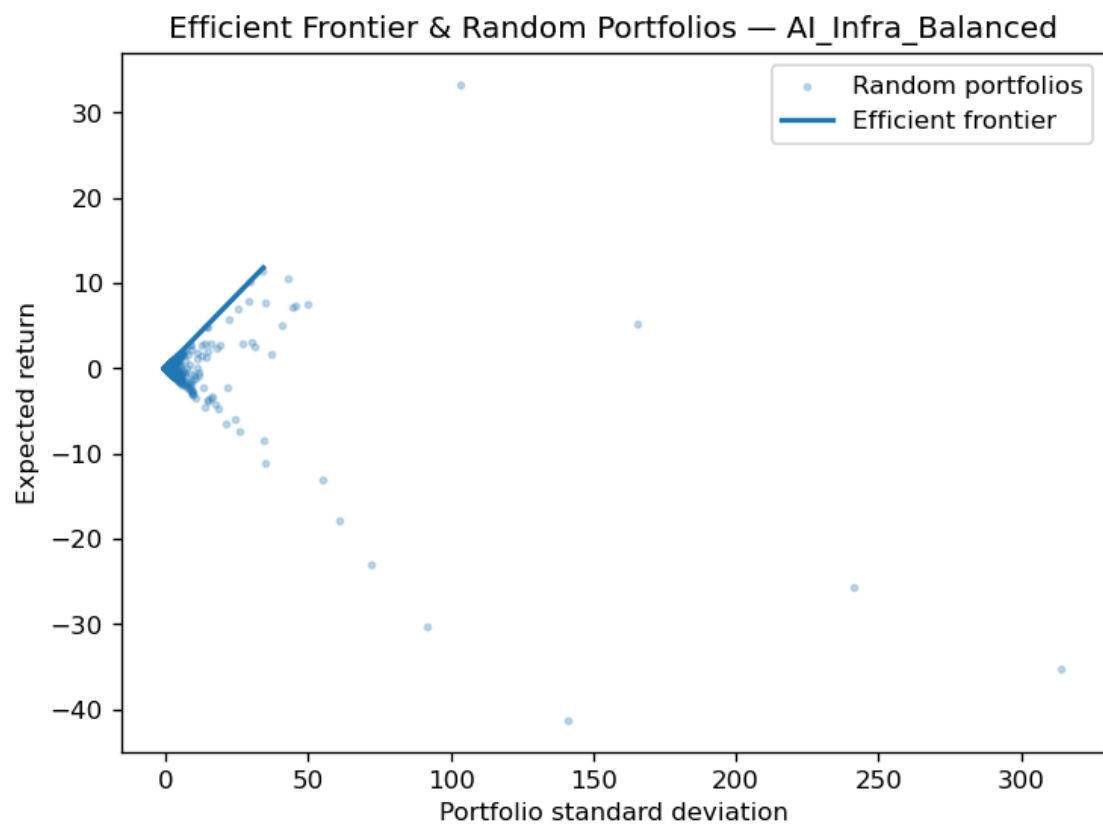


Figure 2: Efficient Frontier & Random Portfolios — AI_Infra_Balanced

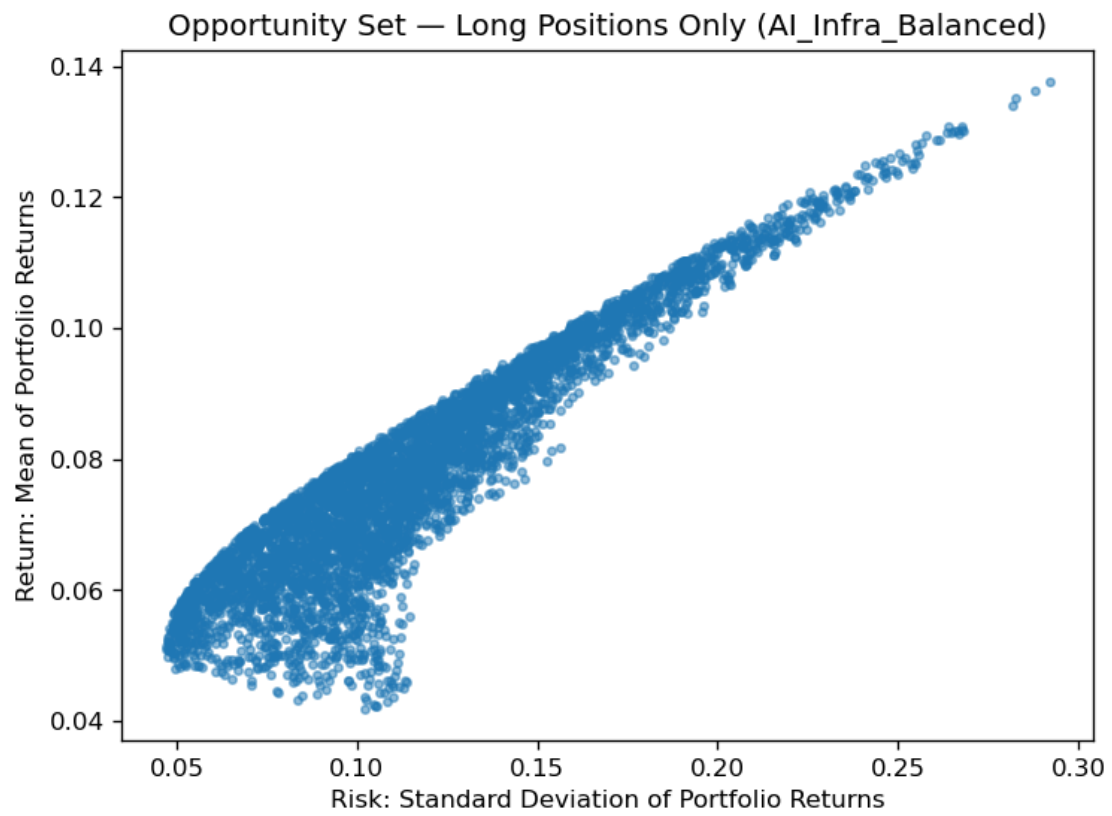


Figure 3: Opportunity Set — Long Positions Only (AI_Infra_Balanced)

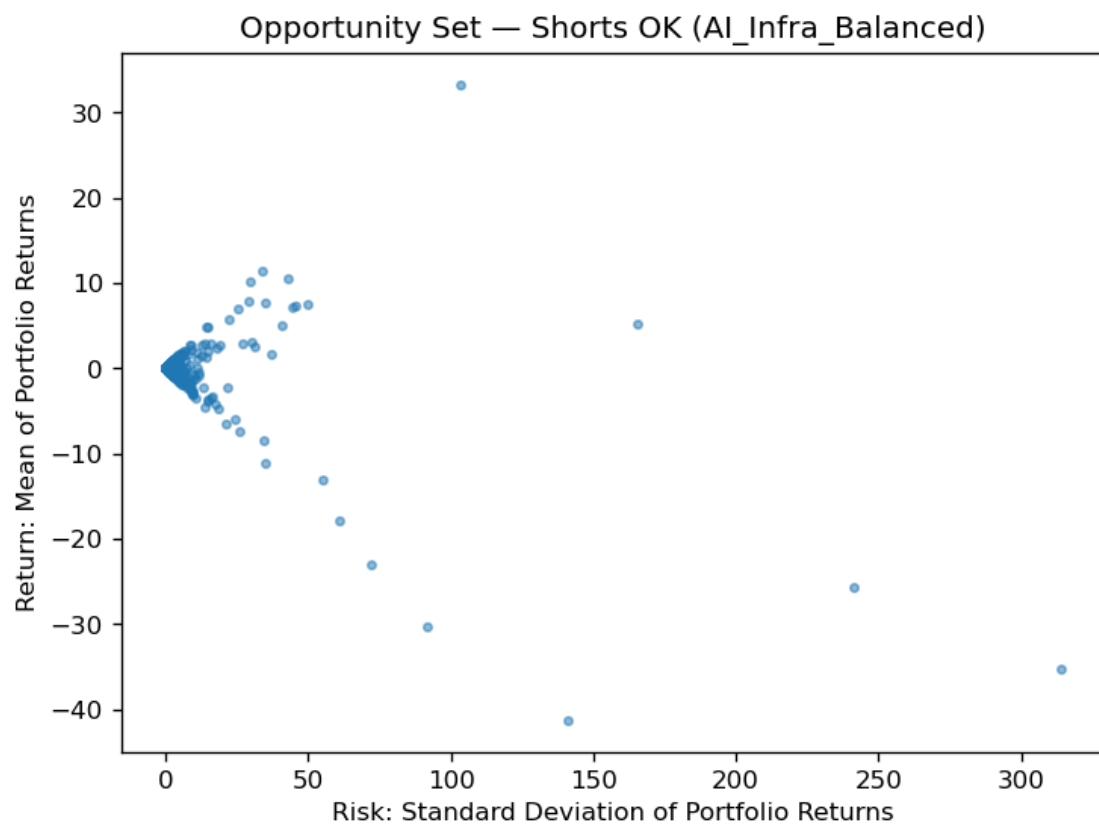


Figure 4: Opportunity Set — Shorts OK (AI_Infra_Balanced)

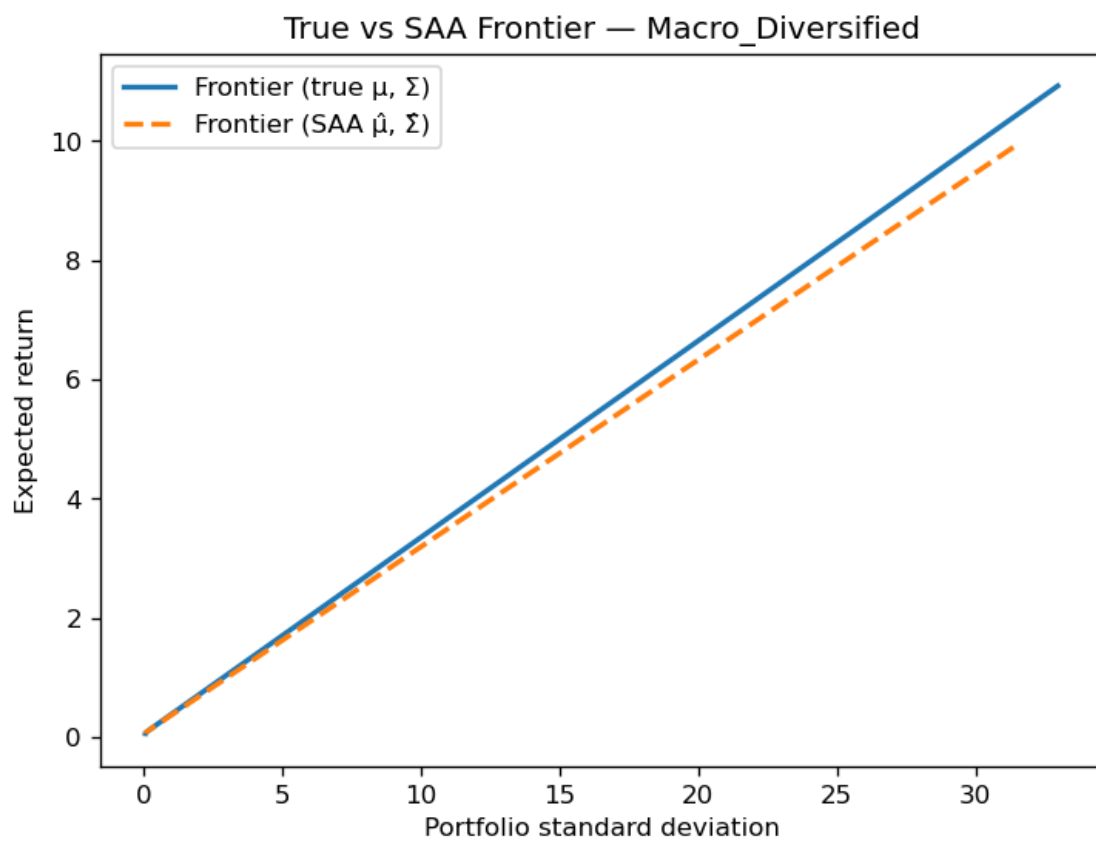


Figure 5: Efficient Frontier (True vs. SAA) — Macro_Diversified

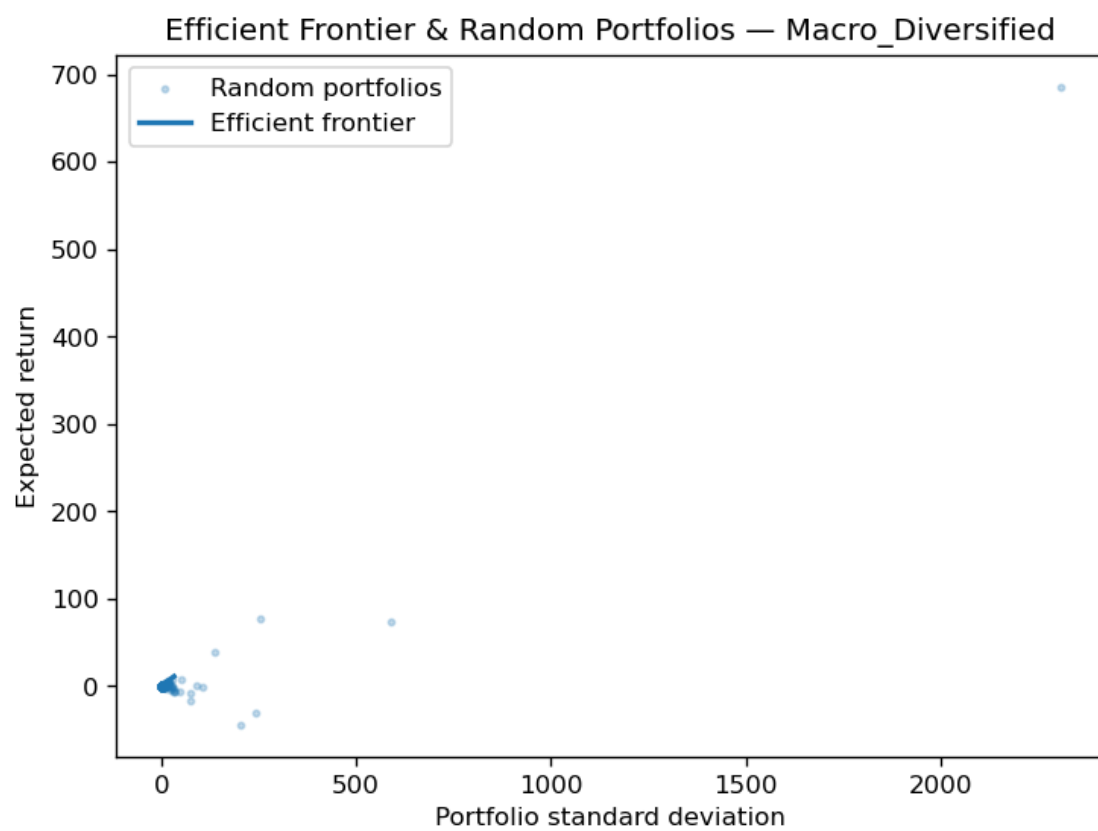


Figure 6: Efficient Frontier & Random Portfolios — Macro_Diversified

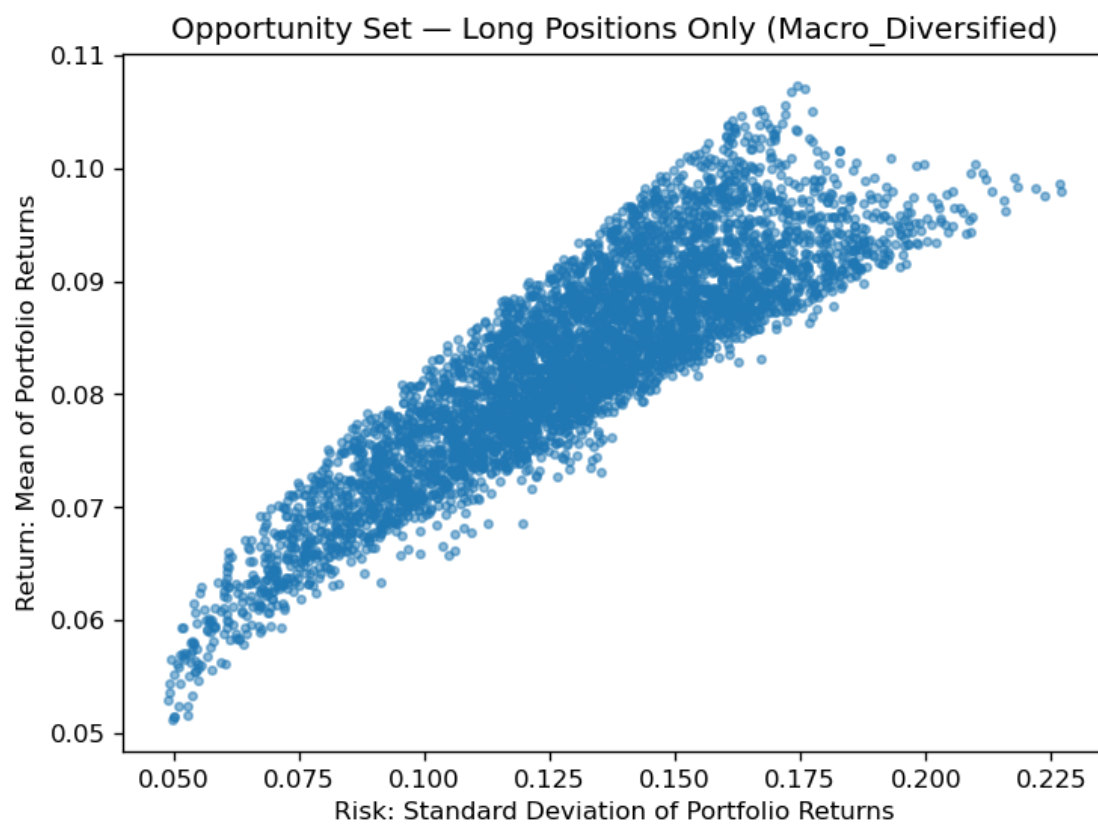


Figure 7: Opportunity Set — Long Positions Only (Macro_Diversified)

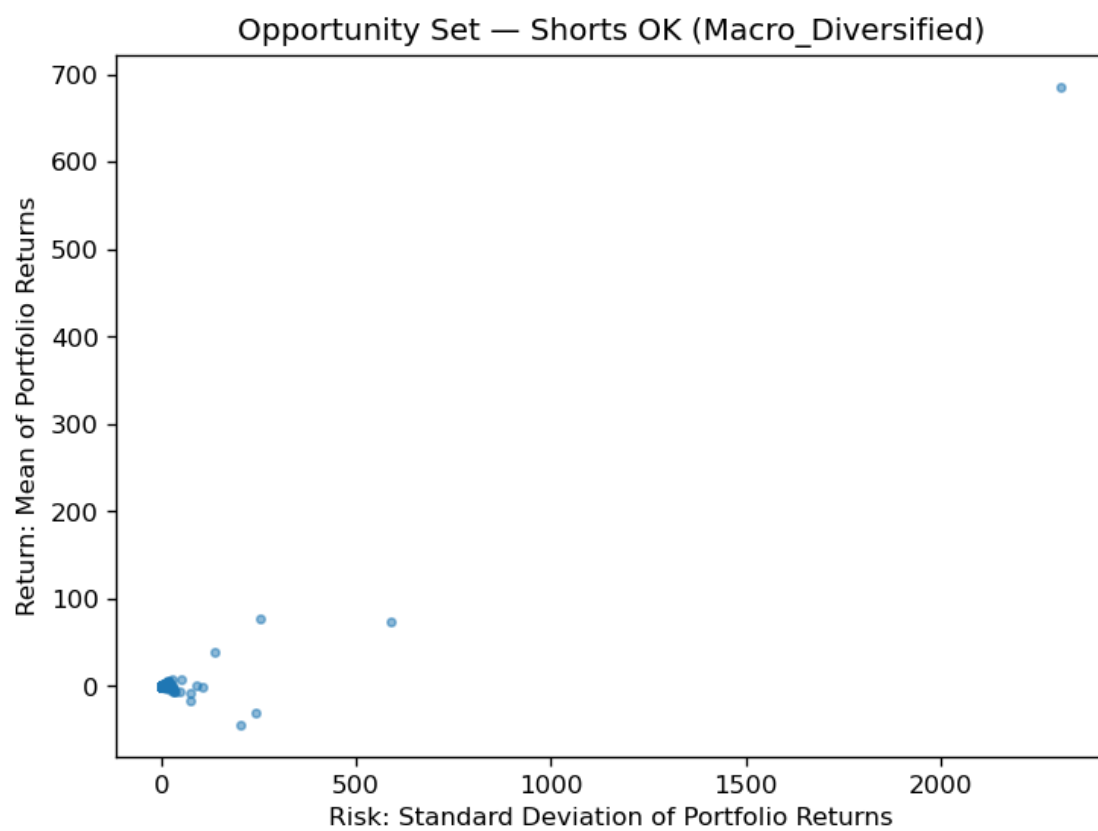


Figure 8: Opportunity Set — Shorts OK (Macro_Diversified)